

Swarm of Drones Using LoRa Flying Ad-Hoc Network

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Abstract—There has been a lot of research to create a robust communication system in a swarm of drones in recent years. With a unique approach towards innovation or optimization, many works focus on extending the coverage but fewer aim decentralization. Long Range (LoRa), a low power modulation technique, combined with Long Range Wide Area Network (LoRaWAN) provides long-distance communication at the cost of throughput. However, the Media Access Control (MAC) level protocol LoRaWAN only supports star topology. This paper contributes towards decentralization by creating a Flying Ad-Hoc Network (FANET) using LoRa in a swarm of drones with a Global Positioning System (GPS) and customized Destination Sequenced Distance Vector (DSDV) routing protocol optimized for a single channel. Results show that load balancing based on hop count is stable, Listen Before Talk (LBT) reduces idle wait time, and dedicated transmitter/receiver improves Packet Delivery Ratio (PDR) and route construction time of LoRa.

Keywords—Flying Ad hoc Networks; Swarm; LoRa; Routing Protocols; Global Positioning System; Drones

I. INTRODUCTION

Ad hoc network in a swarm of drones conventionally has been designed with a short-ranged radio technology such as Wi-Fi (2.4GHz). While long-ranged radio technologies offer greater range, but it results in low data rates. This requires a minimalistic approach to design communication protocol with smaller instruction packets. One such long range technology is Long Range (LoRa) [1], a Sub-GHz radio technology originally developed by Cycleo, later acquired by Semtech.

LoRa is a non-cellular modulation technology which consumes low power and provides long distance communication at low data rates. Long Range Wide Area Network (LoRaWAN) is the MAC level protocol which uses LoRa technology to create a network of star topology. In the world of Internet of Things (IoT), LoRaWAN strives to provide bidirectional secure communication, and is maintained by LoRa Alliance [2]. LoRaWAN consists of two types of entities in a network. Nodes are the first ones, which have some functionality to do such as gather data from an environment and send the result one-hop away to a central entity called a Gateway. Gateways may send the data further to a locally connected database or upload to a cloud server if it has internet access. The communication range is much farther compared to 2.4GHz radios [3].

While LoRa offers long distance communication, but the LoRaWAN protocol has some limitations while considering it for a swarm of drones. First and foremost, it has a single point of failure, which is completely in contrast to a decentralized swarm network. Secondly, ‘after sending an uplink packet, an end device opens receive windows at 1 and 2 second delays for downlink transmissions’ [4, Section III, para. 3]. Swarm needs a real time, reliable and fast communication. A delay

would yield undesirable communication. Thirdly, LoRaWAN is specially designed for IoT applications where a node transmits small amount of data periodically in hours or days to consume less power with 1% duty cycle [5]. But the frequency of data transmission in a swarm will be much greater in comparison, such as in seconds or milliseconds.

Considering a swarm of drones, there are some use cases where a decentralized network is preferred compared to a centralized one. For example, communication with the swarm through any drone at sight as in Fig.1. Any drone which is not at sight, can be communicated via drones at sight acting as relays or multi-hops. Another example would be freedom from the coverage area of gateways. Theoretically, it is possible to place gateways through vast lands in the form of base stations or flying zeppelins, but that would be challenging. Also, during a time-critical disastrous mission, a chain link of drones would be preferred rather than setting up a gateway.

For an ad hoc network to forward packets in a swarm, there must be some collection of protocols which must act dynamically with mobility models [6]. There are different types of routing protocols being proposed for Flying Ad Hoc Network (FANET) [7, 8, 9]. Some are brand new, while others are an optimized version. All the distance vector and link state protocols are categorized into proactive, reactive, and hybrid ones. Distance vector routing protocols keep track of neighbours only while link state routing protocols keep track of all links in a network. Former is like a road sign and latter is like a complete map. When it comes to LoRa, either there must be a new approach or these existing protocols must be optimized to be used with the limitations of LoRa.

The main motivation of this paper was to contribute towards decentralization of swarm of drones using LoRa, Customized Destination Sequenced Distance Vector (DSDV), and Global Positioning System (GPS). The paper is organized as follows: Section II discusses related works. Section III includes design and implementation of proposed approach. Section IV includes results of our approach. Section V includes conclusion and future aspects.

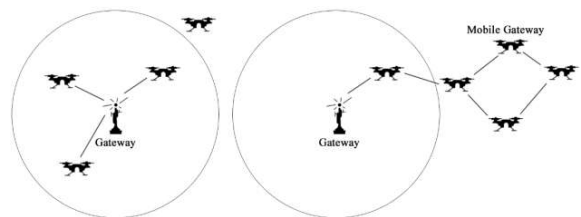


Fig.1. (Left) Centralized swarm. (Right) Decentralized swarm with two gateways, base station and mobile gateway.

II. RELATED WORKS

Researchers have been working on a solution to create a robust communication system in a swarm of drones. These works address different problems with different approaches and methodologies within the domain. R.P. Centelles, F. Freitag, R. Meseguer and L. Navarro in [10], have nicely summed up many of the research works done on LoRa Meshwork and some of its use cases. Future challenges are also being discussed, of which decentralization is the least worked on. We have also looked into most related and relevant works, but the number of works done to integrate LoRa with swarm of drones in the form of ad hoc network are almost negligible if we consider implementation.

Abrado and Pozzebon [11], have created a Linear Sensor Network (LSN) to monitor medieval aqueducts dug beneath the Centre of Siena, Italy. They found out that transmission range of LoRa is limited to 200 metres, so they came up with a chain-link communication network. Their main goal was to prolong network lifetime with low sampling rate. All nodes are evaluated based on wake-up time to minimize energy consumption. Network timing and duty cycling played a vital role in this project. This led to 50% reduction in power consumption. However, there was a problem with fault tolerance, as the chain can break leaving all nodes after break point disconnected. This project had static topology and real implementation of the system is left for further research.

Panicker, Azman, and Kashyap [12], proposed a novel design of a mesh network formed by LoRa based animal collars or tags for animal tracking location. Sensors such as GPS can be embedded onto the collars. They came up with a multi-hop solution instead of placing multiple gateways on a field. Three models were presented. The first is a meshwork of only nodes, second is a meshwork of only gateways, and third is a meshwork of both. The gateway communicates with only one node at a time and has a waiting list for all other nodes, and it gives a certain clock cycles to each connected node. The nature of animal's movement is dynamic, however there is a lack of mention of which routing protocol been used.

Lundell, Hedberg, Nyberg, and Fitzgerald [13, 4], presented a new routing protocol to enable mesh networking with LoRaWAN, allowing for multi-hop networking between gateways to extend coverage without internet backhaul. The protocol is based on Hybrid Wireless Mesh Protocol (HWMP) and Ad Hoc On Demand Distance Vector Routing (AODV) adapted to LoRa networks, which adds extra 44 bytes to already 13 bytes of LoRa message. They created a proof-of-concept and validated in laboratory and field tests. However, in our use case, there is a bidirectional communication. In LoRaWAN, there is a time constraint on the receive windows. After sending an uplink message, an end device opens receive window after 1-2 second delay for downlink transmission. Downlink transmission is not being tested in this paper.

Robert Olsson [14], came up with intelligent gateways as a solution for meshwork where there is no internet backhaul or where shared internet is required. They used a routing protocol based on Ad hoc On Demand Distance Vector (AODV) routing protocol and this solution also helps in coverage extension and downlink messages were also addressed and tested. The solution added only 3 extra bytes for routing to the already 13 bytes of LoRaWAN header size.

Dias and Grilo [15], implemented a new design which provided a solution for coverage extension using multi-hop

uplink. These multi-hops include intermediaries which act as relays from end nodes to gateways. They took DSDV protocol and simplified it for their use on relay nodes. This solution is compatible with existing LoRaWAN, but the downlink was not addressed. The implementation took place using linear and bottleneck topology, where Packet Reception Ratio (PRR) and throughput did not decrease considerably in the linear topology with the increase of hops.

Nelson C. Almeida et al. [16], presented and verified a proposal to create a mesh network of LoRa nodes connected to a proxy node. This proxy node is connected to LoRaWAN network and acts as a coordinator thus, integrating LoRa mesh network with LoRaWAN. The proxy node implements a simplified Ad hoc On Demand Distance Vector (AODV) routing protocol. End nodes are always waiting for a route request to reply to proxy node with a route. If end nodes receive a route request, they reply with another route request from proxy node. If the proxy node does not reply, the current route is valid.

Chahine and Lewin [17], worked on creating a peer-to-peer network to exchange geolocation messages in rural areas using LoRa and GPS modules. They chose Received Signal Strength Indicator (RSSI), Signal-to-noise Ratio (SNR), and Packet Delivery Rate (PDR) as their performance metrics. They did a field test with a distance limit of 1.1 kilometres (km) on a direct Line of Sight (LOS). They used LEE and Hata models to calculate the RSSI, SNR, and PDR. Their results showed that LEE model was more ideal for their project.

Huh and Kim [18], studied LoRa-based private networks and proposed a modified LoRa protocol stack that can replace LoRaWAN. It overcomes the shortcomings of existing LoRaWAN, one of which is private mesh network to improve network coverage and new access schemes. It also supports time-division multiple-access. They found that an imbalance zone exists where end nodes can have easy downlink transmission but uplink transmission has problems. Using their protocol, the uplink can be relayed via multi-hops.

Xiaofan Jiang et al. [19], proposed and demonstrated a hybrid, low-power, wide-area network (LPWAN) structure that uses both sub-GHz long-range and 2.4GHz short-range radios by integrating them with low-power microcontroller. A Mesh network was formed using LoRa with the help of time-division multiple access (TDMA). For data collection they used ANT, which is an ultralow-power, short range communication protocol. Their demonstration included real-world deployment. They came with results suggesting that both networks have superior advantages in terms of cost, coverage and power consumption compared to LoRaWAN.

Hester et al. [20], developed Meshtastic, a project that uses GPS mesh radios with LoRa chip to create a mesh communicator using simple flooding of data. Users create private flat mesh to exchange their location where there is no internet or no centralized network. The batteries need recharging after about eight days. The project is meant for hiking, skiing, paragliding and other outdoor activities. It is currently in beta-testing and completely open source.

The above works do not consider LoRa mesh application with dynamic scenarios where location update is constantly needed, such as swarm of drones. Most of the works consider scenarios where simple mesh network is needed, but in this paper, the Physical Layer payload of LoRa will contain the protocol itself replacing default LoRaWAN.

III. SYSTEM DESIGN

In this section we discuss the methodology and implementation this paper puts forth to create a LoRa flying ad hock network. The experimental setup is given in Fig.2. There are a lot of options to choose from as a base microcontroller/microprocessor ranging from Raspberry Pi to Arduino. A complete operating system is not needed, but rather simple tasks will be carried out such as to store GPS data and routing tables which can be achieved using single thread. Arduino's microcontrollers are easier to interface with sensors compared to Raspberry Pi. Initially, Arduino Uno was considered but 32 KB of memory was less for our work. So, Arduino Wemos R1 D1 with 4 MB of memory is more than enough, and it can be programmed easily using Arduino IDE which uses some functions of C++.

As for LoRa transmission, Dragino LoRa shield v1.4 transceiver was chosen because it is specifically made for Arduino form factor. LoRa Alliance has allocated the frequency range of 864.0-869.2 MHz in Russia, which made us select a frequency of 868.0 MHz. This was interfaced with Wemos R1 D1 as shown in Fig.3. Serial connection between Arduino and the Drone is sufficient for our data, and also U-blox GPS module was connected with the drone. The simple broadcasts in the program were coded in C++ using LoRa library developed by Sandeep et al. [21]. The reason of using this is that this library exposes radio layer directly in form of broadcasts. By default, there is no addressing, encryption and additional layers such as provided by LoRaWAN.

The approach considers both a network with backhaul internet connection and without it. The parameters include the use of Listen Before Talk (LBT), decentralized access to the network via any drone, single-channel (868MHz), custom application layer DSDV protocol which has its own three types of instruction sets and three types of information tables.

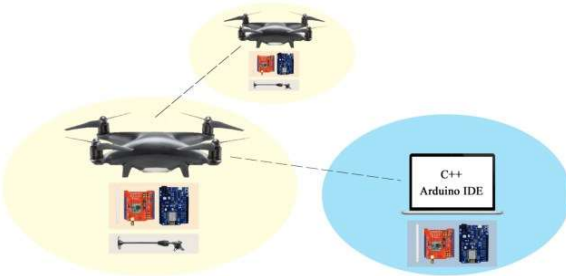


Fig.2. Experimental Setup: Each Drone has Arduino Wemos R1 D1, LoRa Shield v1.4, and U-Blox GPS connected. The Laptop is acting as an access point for commands and also acting as a test node.

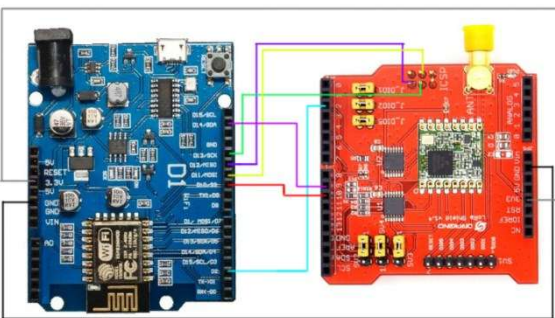


Fig.3. D10 to NSS | D2 to DIO0 | 3.3v to 3.3v | GND to GND | D14 to RESET | D13 to SCK | D12 to MISO | D11 to MOSI

A. Creating the mesh network

For the creation of mesh network, keeping in mind the 13 bytes header size of LoRa, custom DSDV protocol was created at application layer, in other words Payload of LoRa packet. The standard DSDV protocol is around 12 bytes in size, but the one in this paper has three types of instruction sets each with different sizes as in Fig.4. Adjacency Hello Packet (AHP), Routing Hello Packet (RHP), and Command Hello Packet (CHP) are broadcasted periodically. Each drone has a unique Device ID and broadcasts Adjacency Hello Packet (AHP) (12 Bytes) periodically that includes GPS information (with Altitude) and its ID. It also includes information whether a broadcaster has internet access for default routes. The frequency of these AHPs was set to 1 second. Any drone receiving an AHP, creates an entry to its Adjacency Table with ID and GPS location, meaning it's an adjacent neighbour.

The non-adjacent member or indirectly connected member is known through Routing Hello Packet (RHP). Each drone also maintains a Routing Table. After every second, the adjacent entries from Adjacency Table are copied to Routing Table. Then this Routing Table is broadcasted to adjacent neighbours using Routing Hello Packets. A type of packet which shares all entries from Routing Table to adjacent neighbours. The size of this packet depends on whether GPS information is needed. For only hop count, the size is 3 bytes and 1 byte for each route entry including Device ID. When GPS information is needed, the size becomes 10 Bytes, but a separate transmission for each route entry. It could be done in two ways: Broadcast information about each route entry or make a sequence of route entries then broadcast.

CHP is a command level instruction format, which can be sent to any drone. It tells the receiver about a Device ID and a GPS location to go to. This packet is forwarded using AHPs and RHPs. A Forwarding Table acts as a queue to store and execute CHPs one by one. AHPs, RHPs, and CHPs have timeout timers to keep track of them. When the timer runs out, the entry is removed until it receives a new one or updates it.

B. Listen before talk and decentralized access

In a single channel implementation, a problem lies in a shared usage of a channel. This requires a timer/clock that is synchronized throughout all drones. However, it limits scalability by dividing access time of a channel among the drones. Idle devices still use the time slot and also need synchronization. Listen Before Talk (LBT) [22, 23], is an alternative solution without a timer that corresponds to Carrier Sense Multiple Access (CSMA), where we check a channel if it's available before transmission as in Fig.5. But with LoRa, there is a form of delay lasting up to 2 seconds that occurs after an uplink transmission followed by a downlink transmission. To avoid this delay, dedicated transmitters and receivers are needed, where we have an uninterrupted listening to a channel.

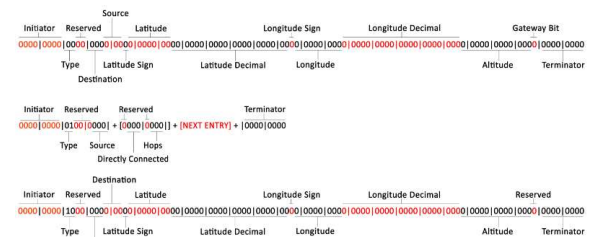


Fig.4. Instruction sets of Adjacency Hello Packet, Routing Hello Packet, Command Hello Packet in the payload.

To further reduce collisions, LBT can be coupled with timer where each drone is assigned a certain time group. In this way, the issue of idle drones in timer/clock approach is solved.

The mesh is decentralized in a flat way, which means the protocol considers drones and base stations as a part of meshwork with the same level of participation and privileges. Any drone having internet access can act as a gateway of last resort if internet access is embedded, otherwise it will forward packets to a route having an internet access. This is achieved with the 'gateway bit' in the instruction format, which helps in keeping track of the gateway. The network also accepts commands from multiple sources, in other words there can be more than one author of the network behind as shown in Fig.5.

C. Distance measurement and load balancing

Hop count is irrelevant for understanding distances when GPS coordinates are given. From point 'A' to point 'B' via many hops, the greater the distances between these hops in a path, a smaller number of hops are required. The smaller the distances, a greater number of hops are required. This is a dynamic environment, where the distances keep changing and rarely two paths become equal in distances between these hops. Considering this, it is a good idea to use hop count for the purpose of Load Balancing as Equal Cost Multi Paths (ECMP) since hops change less frequently as in Fig.5.

Distance calculated using 'Haversine' formula does not consider altitude, but for this paper calculating the great-circle distance is enough. Straight-line distances may pass through the Earth if the distance is too long, but a distance of LoRa transmission will unlikely pass-through Earth. The Haversine formula is given in (1), where ' Δ ' is difference, ' Φ ' is latitude, ' λ ' is longitude, and 'R' is radius of the Earth (6371 km).

D. Proactive distance vector updates and scalability

The three types of instruction sets are broadcasted proactively in the form of Hexadecimal string. There is a way to further decrease the size of string to half by introducing a custom numbering system using symbols. In that case, one symbol will represent 8 bits unlike hexadecimal where one symbol represents 4 bits. This needs a lookup table to be created to search the corresponding 8-bit pattern against the symbol. For now, the instruction set is not encoded since our focus is on decentralization.

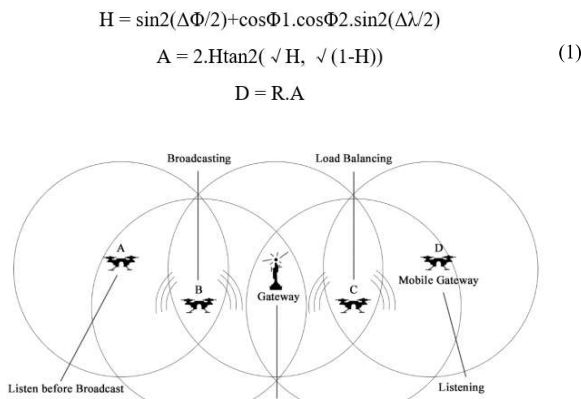


Fig.5. 'A' is listening and waiting for 'B' before broadcasting. 'C' is load balancing between two gateways with same hop distance. Both gateways are authors of the network.

Reactive protocol in this environment needs checking validity of routes, when instruction is executed, because of dynamic nature. Each update requires route discovery from the beginning. Each route discovery needs a two-way confirmation for the route. First is the discovery of route from source to destination, and second is the confirmation of route from destination to source. There is also another method of simply flooding the whole network. It is a robust way but it creates a broadcast storm and energy wasting avalanche. To avoid this, proactive protocol is used in this paper.

Although, DSDV is a proactive protocol, at least it is not a link state routing protocol. It has extra overhead because of table sharing to immediate neighbours. But compared to link state routing protocol information, which keeps link state information of the whole network, DSDV table information is very little for small private networks. As the network increases in the number of drones, DSDV becomes less scalable. To make it more scalable, this requires zonal division of the network. Between two zones, drones with new roles can be introduced which keep summary routes. This is discussed in the future aspect in this paper.

IV. RESULTS AND FIELD TESTS

A lab and field test were done to test the working of what this paper proposed. The lab test did not include simulation software, but rather every parameter was observed manually on console by programming. A LoRa node was connected to the computer and lab results were serially monitored. The field area was in proximity to Tomsk State University. One gateway and two drones (gyrocopters) were tested. Our calculation matrix includes load balancing using hop count compared to GPS distances, LBT compared to Timer, Packet Delivery Rate (PDR) with respect to range, route construction time.

A. Load balancing

We gave two drones the status of mobile gateways, and the base station (where we were) was to do load balancing between these mobile gateways. ECMP was more stable and more area encompassing using hop count instead of GPS distance as shown in Fig.6. The dark circle shows the path of drones, and red spots show where load balancing was active. Remember, ECMP requires path costs to be the same.

B. Listen before talk and timer

We allocated 500 milliseconds for each drone to broadcast. Drone 3 was set to idle, without any broadcasts but only with a time slot. It can be seen that idle drones waste the time division in clock timer method, but in the LBT method these idle slots can be used by other drones as seen in Fig.7. Keeping both methods in mind, the more the number of drones, the greater is the gap of time for each drone.

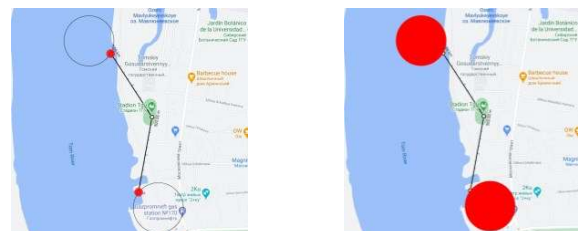


Fig.6. (Left) Using GPS Distance. (Right) Using Hop Count.

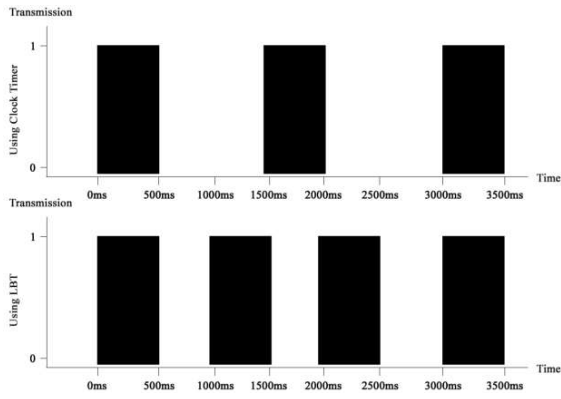


Fig.7. Transmission of Drone '1' using timer compared to Listen Before Talk. At 1000ms, its Drone '3' turn to transmit but it is idle. Drone '1' using LBT to check the channel, is able to transmit during this time.

C. Packet delivery rate and range

PDR is the ratio of packets successfully received compared to the packets sent. For this, we created a 2-hop topology as shown in Fig.8. We compare the PDR of first hop to the range increase and PDR of second hop to the range increase. For the second hop, we also consider and add max distance of 1000 m from the first hop. Lab tests are also given as in Table 1. It can be seen that as the range and hops increase, the PDR starts decreasing.



Fig.8. Adjacency and Routing Table of Drone '1' with only one adjacent neighbor Drone '2' and a route to non-adjacent member Drone '3'.

TABLE I. PDR to Range calculated at Drone 1 of 1000 Packets.

Range in Meters (m)	Packet Delivery Ratio (PDR)				
	First Hop		Second Hop		
	Lab Test	Field Test	Lab Test	Field Test	
200	0.9834	0.9711	1000+200	0.9744	0.9516
300	0.9764	0.9675	1000+300	0.9613	0.9489
400	0.9607	0.9610	1000+400	0.9535	0.9403
500	0.9531	0.9550	1000+500	0.9415	0.9350
600	0.9446	0.9399	1000+600	0.9314	0.9222
700	0.9366	0.9266	1000+700	0.9281	0.9149
800	0.9301	0.9201	1000+800	0.9212	0.9021
900	0.9213	0.9113	1000+900	0.9110	0.8967
1000	0.9145	0.9045	1000+1000	0.9049	0.8838

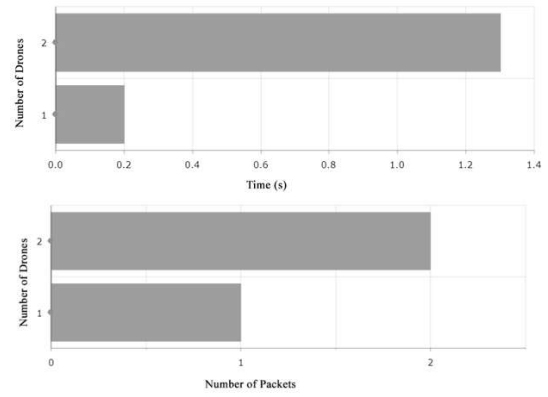


Fig.9. Route construction time and number of packets needed to construct them.

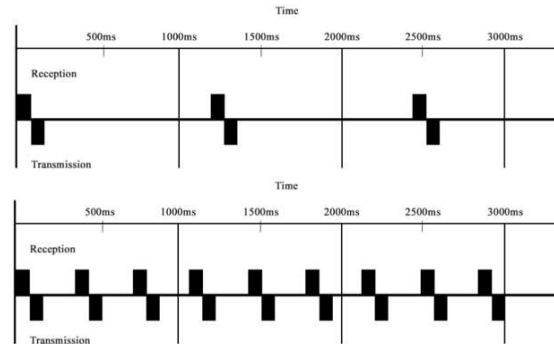


Fig.10. Single transceiver compared to dedicated transmitter and receiver.

D. Route construction time and Delay

We calculated routes construction time and the number of packets required to complete route construction as shown in the Fig.9. The route construction time is required to create routes to all drones in a network. In our case, the construction time was for 2 Drones.

Delay when using single transceiver compared to dedicated transmitter and receiver showed significant difference. We sent an acknowledgement message back to every AHP received for testing the 1-2 seconds LoRa transmission delay as mentioned in [4, Section III, para. 3]. For every AHP reception, there was immediate transmission back. Results are given in Fig.10. We tested this in a lab environment by broadcasting AHP every 300 milliseconds.

V. CONCLUSION AND FUTURE ASPECTS

A. Conclusion

This paper proposes, discusses and implements the integration of LoRa technology with a swarm of drones. Some traditional computer networking concepts are being merged into FANET, such as load balancing and carrier sense, in order to create a decentralized meshwork with field and lab tests. The final protocol aims decentralization in single channel low-powered long-ranged wireless technologies in FANET, by introducing an effective protocol selection in scenarios where load division and listening before on-demand broadcast is required to avoid collisions. This is achieved by introducing LBT solely or by combining it with time division, and by basing load balancing on hop-count.

DSDV was modified with custom instruction sets to make it compatible for low data rates, which included GPS position information and broadcasted. Load balancing was introduced with complete decentralized access where load balance considered equal paths, and a new alternative to time division was introduced. Our results showed that load balancing based on hop-count had significant stability compared to GPS distance. Using LBT instead of timer helped to avoid wasting time which usually happened in case of idle drones. Our instruction format showed a very good PDR for low-powered long-distance communication with a very good route construction time.

B. Future Aspects

The project implements and shows a glimpse of implementing low-powered long-ranged radio technology LoRa with a swarm. Because of time limit and harsh weather conditions, we could not look into some improvements we were also considering. But we decided to include them in the next paper which is ongoing.

The first improvement was to introduce area system in the network. Just like Open Shortest Path First (OSPF) in computer networks, a simplified version can be introduced with Area Border Drones (ABDs) which do route summarizations. This will greatly improve Scalability.

Another improvement was to integrate a private Block Chain with the swarm of drones for security. As Blockchain is a distributed ledger, the idea is to create a ledger of commands history for authentication and validation of authors behind the network. If there is any intrusion to alter the data within the blockchain of a drone, it must be validated by all other drones in the network otherwise it will be rejected.

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